Ex.no:7b	DATA VISUALIZATION USING TABLEAU DIABETES PREDICTION DATASET

Aim

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective data analytics using the Diabetes Prediction Dataset.

Algorithm: Diabetes Prediction Visualization using Tableau

Step 1: Data Loading

- Connect the Diabetes Prediction Dataset to Tableau.
- Import the diabetes_prediction_dataset.csv file.
- Verify the imported fields: gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level, blood_glucose_level, and diabetes.
- Verify that the numerical fields are recognized correctly as numerical measures and categorical fields are recognized as dimensions.

Step 2: Blood Glucose Bar Chart

- Create a worksheet for gender-wise blood glucose analysis.
- Place gender on Columns.
- Place blood_glucose_level on Rows.
- Aggregate blood_glucose_level using Average.
- Select the Bar Chart visualization.
- Enable data labels to display the average blood glucose level for each gender.
- Use the chart to compare blood glucose levels between gender categories.

Step 3: Smoking History-wise Diabetes Visualization

- Create a worksheet using smoking_history and diabetes.
- Aggregate diabetes using SUM.
- Visualize the contribution of different smoking-history categories using a Pie Chart.
- Display smoking-history categories and corresponding diabetes cases as labels.

- Use the visualization to compare diabetes cases among different smoking-history categories.

Step 4: Blood Glucose Trend

- Create a Line Chart using age and blood_glucose_level.
- Place age on Columns and blood_glucose_level on Rows.
- Aggregate blood_glucose_level using Average.
- Enable labels to display the average blood glucose level.
- Use the chart to observe the variation in average blood glucose levels across different ages.

Step 5: Diabetes Percentage

- Create a calculated field named Diabetes Percentage.
- Use the formula:

SUM([diabetes]) / COUNT([diabetes]) * 100

- Display gender, diabetes, and Diabetes Percentage in a worksheet.
- Visualize the diabetes percentage for each gender using a Bar Chart.
- Enable labels to display the percentage values.

Step 6: Smoking History Filter

- Add smoking_history to the Filters shelf.
- Select a smoking-history category such as never, current, former, or not current.
- Observe the updated visualizations based on the selected smoking-history category.
- Use the filter to compare diabetes-related measurements for different smoking-history groups.

Step 7: Minimum Blood Glucose Parameter

- Create a parameter named Minimum Blood Glucose.
- Set the required minimum blood glucose value.
- Create a calculated field to display only patients whose blood glucose level meets or exceeds the selected value.
- Apply the calculated field as a filter.
- Display the parameter control on the worksheet.

- For example, when Minimum Blood Glucose = 100, display patients having a blood glucose level of 100 or above.

Step 8: Dashboard Creation

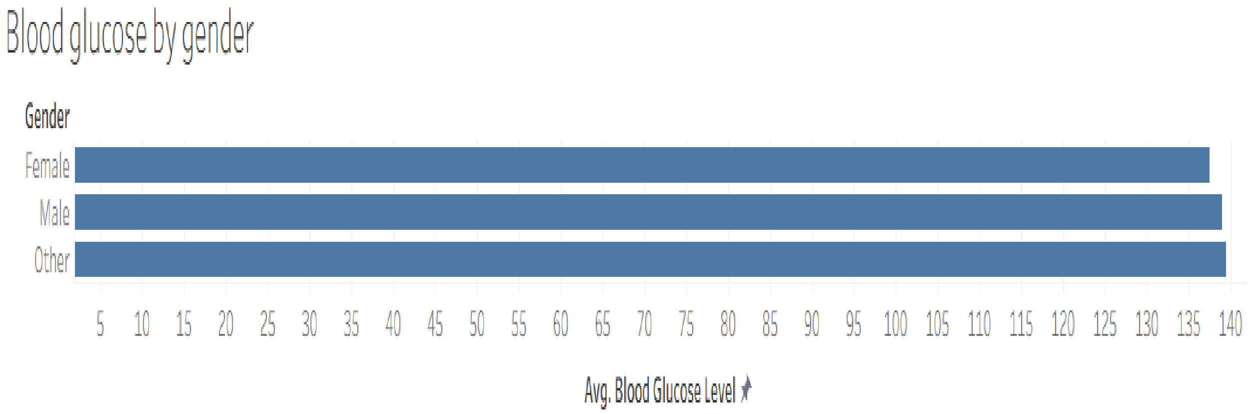
- Create a new Tableau Dashboard.
- Add the Blood Glucose by Gender, Diabetes Cases by Smoking History, Blood Glucose Trend, Diabetes Percentage, and Minimum Blood Glucose visualizations.
- Arrange the worksheets clearly.
- Add the Smoking History filter and Minimum Blood Glucose parameter.
- Format the dashboard with suitable titles and labels.

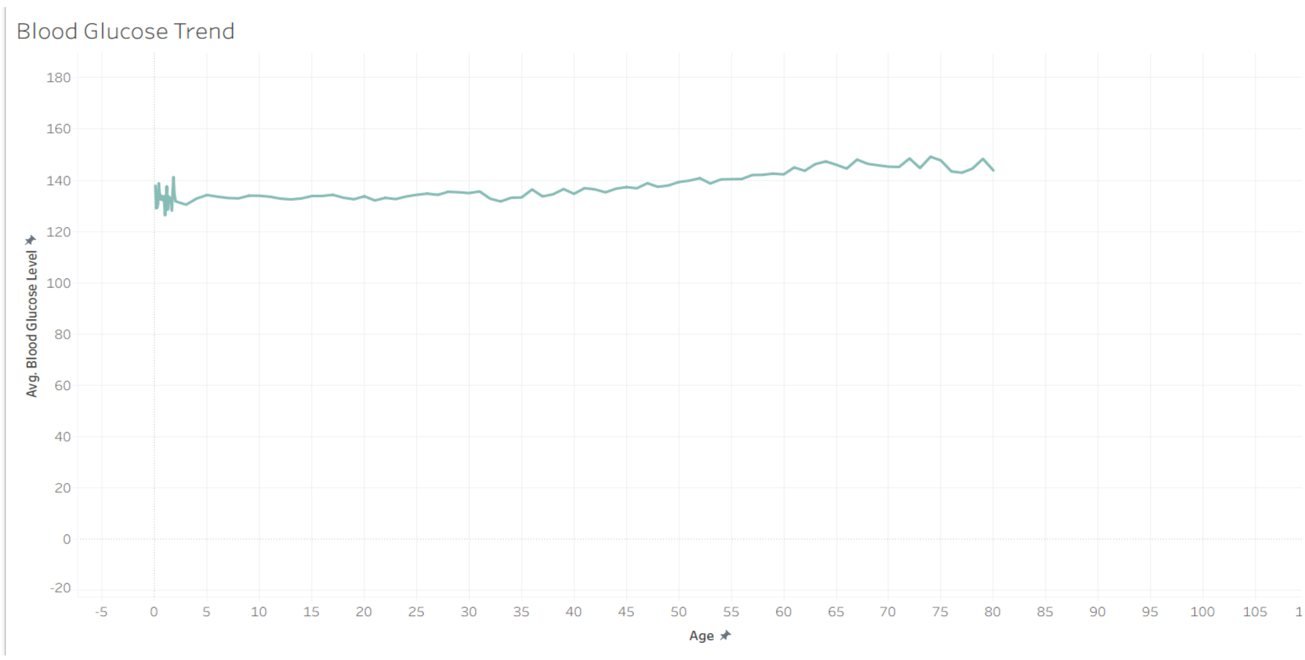
Step 9: Story Creation

- Create a Tableau Story using the completed dashboard.
- Add story points for:
    1. Overall Diabetes Analysis
    2. Gender-wise Analysis
    3. Smoking History Analysis
    4. Blood Glucose Analysis
    5. Final Diabetes Analysis
- Present the visual insights using the story.

Step 10: Analysis

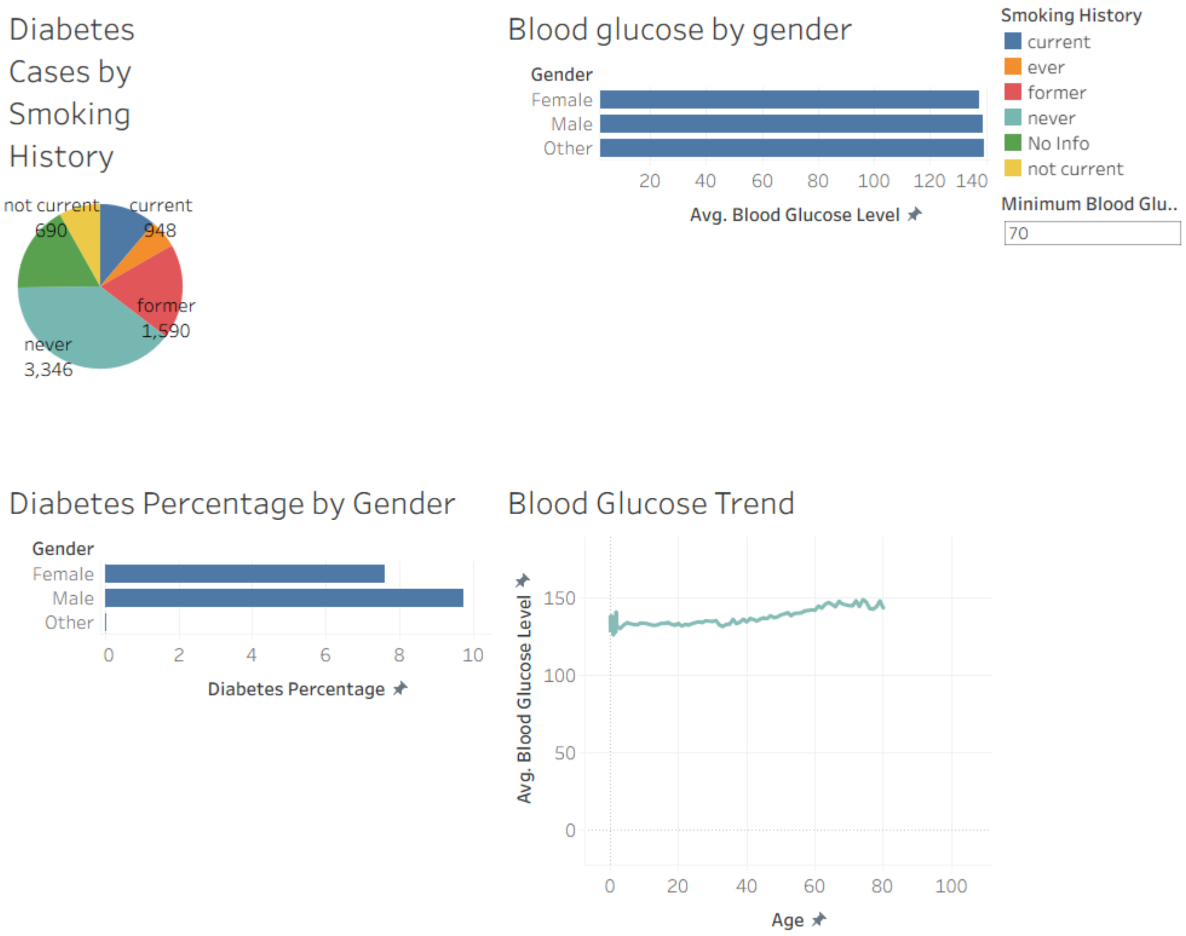
- Analyze the charts to identify the gender with the highest and lowest average blood glucose level.
- Compare diabetes percentages among genders.
- Compare diabetes cases among different smoking-history categories.
- Observe the variation in blood glucose levels across different ages.
- Observe how the Smoking History filter and Minimum Blood Glucose parameter dynamically change the visualizations.
- Use the dashboard and story to summarize the overall diabetes-related patterns.


Blood glucose by gender
Gender
Female
Male
Other
5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100 105 110 115 120 125 130 135 140
Avg. Blood Glucose Level


Blood Glucose Trend
180
160
140
120
100
80
60
40
20
0
-20
Avg. Blood Glucose Level
-5 0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100 105 1
Age

Output

The individual Tableau visualizations and interactive dashboard are created successfully.



Result